

Artificial Intelligence – Exam Answers

Q1. What is Intelligence? Define Artificial Intelligence.

Intelligence:

Intelligence is the ability of an entity to acquire knowledge, learn from experience, understand information, reason logically, solve problems, and adapt to new situations.

Key characteristics of intelligence: - Learning from experience - Reasoning and decision making - Problem solving - Understanding environment - Adaptation

Example: Humans learn from mistakes and improve their decisions.

Artificial Intelligence (AI):

Artificial Intelligence is a branch of computer science that enables machines to perform tasks that normally require human intelligence such as learning, reasoning, problem-solving, perception, and decision-making.

Definition: Artificial Intelligence is the simulation of human intelligence in machines that are programmed to think and act like humans.

Example: Chatbots, self-driving cars, recommendation systems.

Q2. Explain the four approaches to Artificial Intelligence.

There are four approaches to AI based on thinking and acting:

1. Thinking Humanly (Cognitive Modeling Approach)

This approach focuses on making machines think like humans.

Goal: Simulate human thought process.

Example: Neural networks that mimic the human brain.

Advantages: - Helps understand human thinking - Useful in cognitive science

Disadvantages: - Human thinking is complex and difficult to model

2. Acting Humanly (Turing Test Approach)

This approach focuses on making machines behave like humans.

Goal: Machine should act like human.

Example: Chatbots like ChatGPT.

Turing Test: If a human cannot distinguish between machine and human, machine is intelligent.

Advantages: - Practical approach

Disadvantages: - Does not focus on reasoning, only behavior

3. Thinking Rationally (Logical Thinking Approach)

This approach focuses on making machines think logically and correctly.

Goal: Use logical reasoning to make decisions.

Example: Expert systems using rules.

Advantages: - Provides correct logical decisions

Disadvantages: - Real world problems are complex

4. Acting Rationally (Rational Agent Approach)

This approach focuses on making machines act rationally to achieve goals.

Goal: Take best possible action.

Example: Self-driving cars choosing safest path.

Advantages: - Most practical approach

Disadvantages: - Requires complex algorithms

Comparison Table:

Approach	Focus	Example
Thinking Humanly	Think like humans	Neural networks
Acting Humanly	Act like humans	Chatbots
Thinking Rationally	Think logically	Expert systems
Acting Rationally	Act intelligently	Self-driving cars

Q3. Components of AI and Applications

Components of AI:

1. Learning Ability to learn from data.

Example: Machine learning

1. Reasoning Ability to make decisions.

Example: Expert systems

1. Problem Solving Finding solutions.

Example: Chess playing AI

1. Perception Understanding environment using sensors.

Example: Image recognition

1. Natural Language Processing (NLP) Understanding human language.

Example: Chatbots

1. Knowledge Representation Storing information.

Example: Databases

Applications of AI:

1. Healthcare Diagnosis, medical imaging
2. Education Smart tutors
3. Robotics Industrial robots

4. Self-driving cars Autonomous vehicles
 5. Banking Fraud detection
 6. Gaming Chess, video games
 7. E-commerce Recommendation systems
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Current Trends in AI:

1. Generative AI (ChatGPT, image generation)
 2. Autonomous vehicles
 3. AI in healthcare
 4. Robotics automation
 5. Natural Language Processing
 6. Computer vision
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Q4. What is PEAS description?

PEAS stands for:

- P – Performance measure
- E – Environment
- A – Actuators
- S – Sensors

It is used to define an intelligent agent.

PEAS for Cleaning Robot:

Performance Measure: - Clean floor - Less time - Less energy

Environment: - House, rooms, furniture

Actuators: - Wheels - Vacuum cleaner

Sensors: - Dirt sensor - Camera - Obstacle sensor

Q5. Define Intelligent Agent. Explain all AI agents.

Intelligent Agent:

An intelligent agent is an entity that perceives its environment using sensors and acts upon it using actuators to achieve goals.

Example: Robot, self-driving car.

Types of AI Agents:

1. Simple Reflex Agent Acts based on current perception.

Example: Automatic door

1. Model-Based Agent Uses internal model.

Example: Robot vacuum

1. Goal-Based Agent Acts to achieve goals.

Example: Navigation system

1. Utility-Based Agent Chooses best action.

Example: Self-driving car

1. Learning Agent Learns from experience.

Example: Recommendation system

Q6. Four ways to measure performance of AI algorithm

1. Accuracy How correct the results are.

Formula: $\text{Accuracy} = \frac{\text{Correct predictions}}{\text{Total predictions}}$

1. Speed How fast algorithm works.
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1. Efficiency Uses minimum resources.
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1. Robustness Ability to handle errors and unexpected inputs.

Example: Face recognition system measured by accuracy, speed, efficiency, robustness.

END OF ANSWERS

You can write these answers directly in exam.